Gates Foundation

LIGHT TOUCH STRAIGHTFORWARD GRANT – INVESTMENT DOCUMENT

Document Guidance

Welcome! The Investment Document is both your proposal shaping document, as well as your reporting document should your project receive funding. Before beginning, discuss the requirements for this document with your foundation contact for guidance and expectations on content, length, and level of detail.

Please note that based on our understanding of the scope of the proposed project, it qualifies for the light-touch straightforward grant process. While this project qualifies for this light-touch process, you may have other projects that do not.

Important Guidance:

- This document is not a commitment by the foundation to fund the work. If a grant agreement is executed, the grant proposal narrative (Section A) will be incorporated therein by reference.
- To better understand the restrictions on foundation funding, please review the foundation's standard grant terms and conditions. This is not an exhaustive list and is subject to change. This list is provided for informational purposes only and does not imply an award, agreement, or offer to contract.
- This document must be submitted in English and as a Microsoft Word file. **PDFs and other file types will not be accepted**.
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If you have any questions or need support, please reach out to your foundation contact at the email listed on the next page.

GRANT PROPOSAL

Investment Name	AI-Driven Alert Response System for Outbreak Surveillance in Kenya	**Investment ID**	INV-110654
Investment Owner	George Githinji	**Investment Owner Email**	george.githinji@gatesfoundation.org
Investment Coordinator	Amanda Pruzinskis	**Investment Coordinator Email**	amanda.pruzinskis@gatesfoundation.org

INVESTMENT DETAILS

Proposal Submitted By (name)	Dr. Isaac Ngere	**Proposal Submitted Date**	[31/08/2026]
Estimated Grant Start Date	01 November 2026	**Grant End Date**	31st October 2029
Organization details are pre-filled from our records. Please review and update if needed.			
Organization Legal Name[1]	Washington State University-Global Health Kenya	**Total Requested Amount (USD)**	$1,500,000
Organization Doing Business As/Trade Name[2]	Washington State University - Global Health Kenya	**Total Direct Costs (USD)** Direct & Indirect Cost Guidance	$1,335,968

[1] Legal Name will be used in the agreement and should match the name on the bank account that receives the grant funds (assuming fully executed agreement).
[2] Trade Name or d/b/a ("doing business as") only required if different from Legal Name.

Tax Status (if known and applicable) Tax Status Definitions	Public Charity Equivalent	**Total Indirect Costs (USD)** Direct & Indirect Cost Guidance	$164,032
U.S. Employer Identification Number (EIN) (if applicable)	N525PALP83M	**Indirect Cost Rate (as a % of Total Direct Costs)**	12%
Mailing Address		**Primary Contact Name**	Selestine Njoga
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Charitable Purpose - completed by the foundation

- The following describes the charitable purpose of this project. It is written in a standard format by your foundation contact for publication on the foundation's public website should your project receive funding.

To build an interoperable outbreak surveillance platform and shorten detection-to-decision time, enable early containment and build a scalable, disease-agnostic decision-intelligence platform.

Project Description – completed by the grantee

- In this section, please describe the proposed work.

Title: Building a Real-Time Artificial Intelligence-Driven Alert & Response System (RADAR) for Disease Surveillance and Outbreak Management in Kenya

Abstract

Across Sub-Saharan Africa, public health surveillance and response systems remain fragmented, reactive, and poorly integrated across sectors, weakening decision-making and delaying coordinated outbreak response. In Kenya, multiple digital platforms for human surveillance operate largely in parallel with animal health platforms and environmental monitoring systems, creating persistent delays in human disease detection, data harmonization, event escalation, and coordinated response. Decision-support tools are paper-based and remain separate from outbreak management workflow while with poor traceability, meaning that detection is routinely delinked from response, resulting in extended detection-to-decision intervals and inconsistent escalation for similar outbreak contexts. Recent evidence demonstrates an increasing application of artificial intelligence (AI) and data science to tackle data fragmentation and improve epidemic intelligence, with excellent results. Our proposal seeks to develop and deploy a Real-Time Artificial Intelligence-Driven Alert and Response System (RADAR) embedded within the Kenya National Public Health Institute (KNPHI) Emergency Operations Center's (EOC's) outbreak detection and response workflow. The immediate operational focus of RADAR is threefold; i) development and deployment of an AI-enabled data-fusion and system interoperability layer that ingests and harmonizes existing structured and unstructured surveillance feeds from different sectors, ii) integration of predictive risk-based analytics to enable outbreak forecasting and faster disease detection, and iii) deployment of the digitized Decision-Making Tool for Public Health Emergencies (DMT-PHE) within the national and county EOCs, plus integration of its AI-enabled decision-support layer to the surveillance and response architecture, including linkage with Kenya's surge response mechanisms. The goal is a system architecture that ingests multi-stream signals and analytics, applies risk-scoring and anomaly detection, and channels structured escalation recommendations to EOC staff for efficient and timely disease outbreak detection, escalation, and response coordination. RADAR's intended use cases will cover four disease outbreak categories – i) enteric and diarrheal diseases, ii) viral hemorrhagic fevers, iii) febrile and respiratory illness, and iv) public health emergencies of initially unknown etiology (PHEIUE). The RADAR architecture is designed to be deployed alongside the existing EOC data infrastructure, without need to overhaul or re-invent existing systems. The outcome measures will include: i) improvement in disease outbreak

prediction and detection, ii) measurable reduction in outbreak detection-to-escalation latency; iii) demonstrated improvement in 7-1-7 benchmark compliance; and iv) measurable benchmarks and improvement in agentic-AI use and accuracy in outbreak risk classification. The long-term outcome is a scalable, disease-agnostic public health decision intelligence framework that can be rapidly reconfigured across different outbreak threats and other public health problems, and can be scaled to other settings, establishing Kenya as a regional reference for AI-enabled One Health surveillance.

Overall Approach

Background

Although the global health-security landscape has evolved significantly in recent years, surveillance and outbreak decision-making systems at national and sub-national levels remain fragmented, particularly in low- and middle-income countries. In Kenya, multiple national and external assessments, including the Joint External Evaluation (JEE, 2024), the States Parties Self-Assessment Annual Reporting (SPAR), the WHO Strategic Tool for Assessing Risks (STAR, 2023), and the 2025 Gates-funded AWARE assessment, consistently identified fragmentation and limited interoperability across surveillance systems as a critical gap in early warning and response capacity [4]. The human health sector operates primarily under the IDSR framework, with reporting distributed across KHIS2, e-CHIS, and EBS. Animal and wildlife disease reporting is partially digitized through KABS and Earth Ranger platforms, while environmental and meteorological data are managed independently by the Kenya Meteorological Department (KMD) and the National Drought Management Authority (NDMA). While these systems generate substantial volumes of surveillance data, they operate across heterogeneous data architectures, reporting standards, governance structures, and access protocols, limiting real-time cross-sectoral intelligence generation and coordinated decision-making.

The JEE scored Kenya's early warning surveillance function at level 3, noting that the country has not yet achieved full integration of indicator-based and event-based surveillance and that low event-based surveillance coverage disqualified a higher score; it scored analysis and information sharing at level 3 as well, citing the absence of a cross-sector data-sharing framework and the fact that animal-sector data are shared only quarterly. This fragmentation is not a data-quality curiosity; it is the direct cause of Kenya's detection-to-response latency, and the connection between the two runs through a single mechanism. Because signals from different sectors and systems are never joined into one record of the same event, escalation decisions rely on manual reconstruction, decision-support remains largely paper-based [5], and the JEE separately scored activation and coordination of health personnel during an emergency at level 1, the lowest score in the entire evaluation. The result is that detection is routinely delinked from response, which lengthens the interval between a signal and an effective response and undermines compliance with the 7-1-7 benchmark. Kenya today cannot even measure that interval consistently, because the timestamps that would define it are held in different systems with no shared record of the same event. Decision-making at national and sub-national levels is also hampered by a multiplicity of uncoordinated guidance documents that, coupled with weak analytic capacity, perpetuate an ecosystem that is data-rich but insight-limited: capable of recording outbreaks but not predicting or responding to them efficiently. Strengthening the pathway from detection to coordinated response, and making that pathway measurable, is therefore a central aim of this application.

Even where a signal is detected and verified, Kenya's escalation decisions have historically relied on informal discussion among surveillance officers, institutional knowledge and experience, rather than a structured decision framework. KNPHI addressed part of this gap in March 2026 with the launch of the Decision-Making Tool for Public Health Emergencies (DMT-PHE), a standardized decision-support framework aligned with IHR (2005) Annex 2 and the 7-1-7 benchmarks, which functions as the primary escalation interface within the PHEOC [6]. What DMT-PHE does not yet do is draw automatically on the multiple, still-fragmented data streams described above; it is only as good as the information manually assembled and presented to it. The first part of this application will describe how we will strengthen the data that feeds to the response systems, while the second part will describe how we will strengthen and streamline decision making at the PHEOCs using the DMT-PHE.

RADAR Conceptual Design

Traditional surveillance integration approaches often attempt centralized databases or rigid interoperability standards that are difficult to operationalize across fragmented health systems with heterogeneous data architectures. RADAR reframes this challenge by positioning AI-assisted integration and decision intelligence as an operational layer embedded within existing EOC workflows. Recent evidence demonstrates an increasing application of AI and data science to tackle data fragmentation and epidemic intelligence, and the future of healthcare is increasingly becoming AI dominated [7]. The RADAR framework introduces three structured and interconnected layers as follows: (**Figure 1**)

- **Multi-stream data integration layer** consists of engines and algorithms that will ingest and harmonize data streams from human health notifications (IDSR, EBS), livestock and zoonotic disease reporting (KABS), wildlife syndromes (KABS and Earth Ranger), environmental and climate indicators (NDMA and KMD), and contextual datasets such as mobility or behavioral signals derived from telecommunications and digital sources. Using machine-learning approaches, including natural language processing and graph-based models, the RADAR system will learn relationships across datasets, enabling semantic translation, anomaly detection, and early signal fusion across domains to generate surveillance insights.
- **A decision-support layer**[8] is operationalized through the DMT-PHE framework, which incorporates predefined trigger thresholds and escalation pathways aligned with IHR (2005) Annex 2, and Kenya's 7-1-7 benchmarks.
- **An agentic-AI decision layer**[9] is applied to the DMT-PHE interphase to improve functionality and outbreak workflow by further compressing the detection-action loop and improving user experience and accuracy of decision making among EOC staff.
- **A bridge interphase** composed of a minimum event record to track single outbreak events across surveillance systems. This will enable achievement of 'systems integration' ahead of the heavy computing infrastructure needed for full integration.

Kenya's surveillance ecosystem is expansive and increasingly digitized, spanning human, animal, environmental, and social sectors, but each system carries its own data separately, and events that arise across more than one system, which is where the most dangerous outbreaks originate, are the ones most likely to be missed or reported late [10]. When a suspected outbreak occurs in Kenya today, the facts about that single event are scattered. The community signal sits in *m-Dharura*. The animal signal sits in KABS. The case count sits in KHIS2. The laboratory result sits in a laboratory information system. The response decision sits in an EOC spreadsheet or a paper log. No system holds a single record of the event itself.

The operational consequence is that nobody can say, without manual reconstruction, when the event emerged, when it was detected, when it was notified, or when the response began, which is why 7-1-7 performance in Kenya is currently assessed retrospectively and case by case rather than measured continuously [11]. We will support KNPHI to define, adopt, and code a shared minimum event record: a small, standardized set of fields that describe a priority public health risk and the times at which it passed through detection, notification, investigation, and response. The event record is a data standard and an exchange profile in a federated system. It is not a new database, not a new user interface, and not a system that any institution is required to work inside. Existing systems continue to operate exactly as they do now and remain the authoritative source for their own data. What changes is that each participating system can publish a small number of standard fields about an event and subscribe to what others have published about the same event, so that for the first time a single event has a single identity across the human, animal, laboratory, and response systems that touch it. This is the principle upon which the initial bridge version of RADAR will be built. To ensure extensibility and adaptability, RADAR is built on open-standard integration protocols, including the FHIR R4 data standards and Model Context Protocol (MCP), that allow new surveillance data streams to be added incrementally without rebuilding the core architecture.

DMT-PHE decision support layer and the agentic AI version

At the core of the RADAR architecture is an AI-enabled digitized DMT-PHE tool that leverages an expert-curated knowledge base - including materials derived from the paper-based DMT-PHE user guides launched by KNPHI, MOH approved national and international outbreak response guidelines, and relevant scientific literature. Incoming surveillance signals will be channeled through standardized risk-assessment workflows, predetermined thresholds and the AI layer to support surveillance officers by generating structured recommendations for event classification, escalation, and response actions. All decisions, whether confirmed or overridden, are logged in an auditable, time-stamped record. To prevent model hallucinations, the toolkit provides citations to specific documents and pages supporting its reasoning, alongside a transparent reasoning trace. The system maintains a human-in-the-loop model in which the responsible EOC staff confirms or overrides each recommendation, with justification logged. The DMT-PHE toolkit was launched in March 2026 and has been deployed by KNPHI [12]. A digitized version of the DMT-PHE, plus an agentic AI version, developed by our consortium, are already incorporated into RADAR, and has been minimally evaluated for basic performance metrics. ***(Figure 2)***

RADAR operational use cases

RADAR's immediate use case will be in outbreak detection, escalation, and response coordination. The system is intentionally disease-agnostic in architecture but covers four high-impact outbreak categories, each using the same detection-escalation workflow: i) enteric and diarrheal pathogens, including cholera, typhoid, *Escherichia* coli and other waterborne disease

threats; ii) viral hemorrhagic fevers, including Rift Valley fever, Bundibugyo Ebola Virus Disease (B-EBOV), Marburg, and Yellow Fever; iii) febrile and respiratory illness outbreaks, including influenza, mpox, and other systemic syndromes of concern; and iv) public health emergencies of initially unknown etiology (PHEIUE), where early multi-stream signal fusion is critical to accelerating hypothesis generation and escalation decisions before a causative pathogen is confirmed. Should funding and deployment timelines align with an active high-consequence outbreak - such as the current B-EBOV situation in East Africa - RADAR can be brought to immediate operational readiness in direct support of that response. When no acute outbreak is underway, the same architecture supports continuous integration and monitoring of endemic and outbreak-prone diseases, improving preparedness.

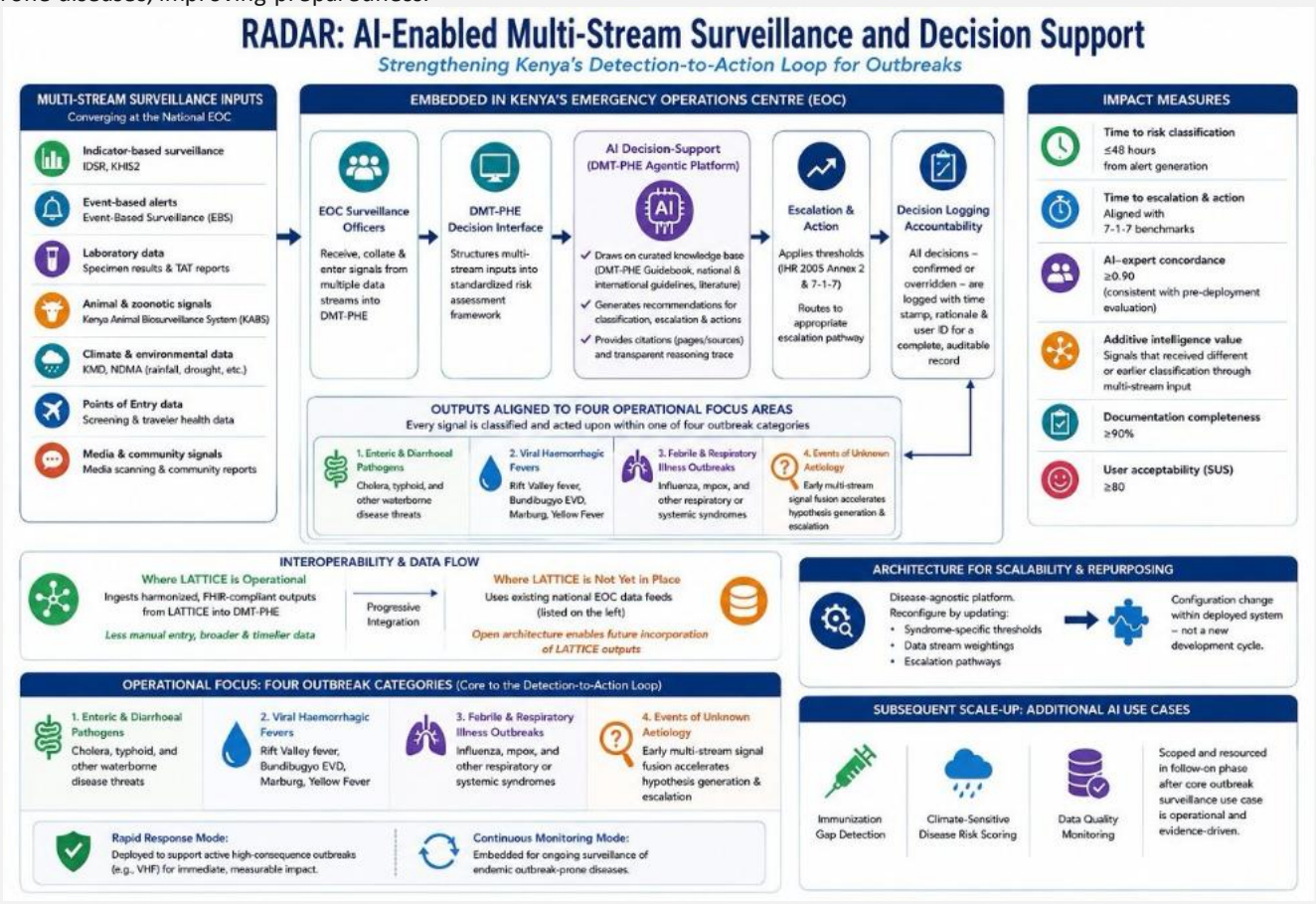


Figure 1*: RADAR multi-stream surveillance and decision support framework. The multi-stream data integration, EOC-embedded DMT-PHE and detection-to-action loops for active outbreak management and passive monitoring of outbreak risks*

Future scalability and repurposing

The RADAR system is designed to operate within the KNPHI-EOC data architecture as a disease-agnostic, reusable public health intelligence and decision support platform. Once operational, the same multi-stream input architecture, EOC-embedded DMT-PHE workflow, and AI decision-support layer can be reconfigured across outbreak categories by updating syndrome-specific trigger thresholds, relevant data stream weightings, and escalation pathway parameters, without rebuilding the underlying infrastructure. This repurposing is a configuration change within an already-deployed system, not a new development cycle.

Preliminary work and background data

To lay the groundwork for this project, the consortium has mapped all data systems that will be integrated into the RADAR platform, together with the KNPHI. We have also developed and evaluated the prototype AI-enabled DMT-PHE workflow. We conducted a facilitated preliminary evaluation of the DMT with 11 surveillance professionals in March 2026 which confirmed deployment readiness: a mean concordance score of 0.924 against the DMT-PHE reference standard, System Usability Scale (SUS) score of 85.2 (Excellent grade), and no true adverse decision risks identified across 33 simulated interactions spanning three outbreak scenarios (*in house evaluation data*). This evaluation confirms alignment of the agentic AI prototype to its intended purpose and provides direct evidence for readiness for national deployment.

Phased implementation steps

The project will be implemented in a three-phase gated approach, moving directly to implementation and product delivery. Each phase (roughly 1 year) has predefined performance thresholds that must be met before the subsequent phase is activated, and has embedded learning agenda, to inform subsequent roll-out stages.

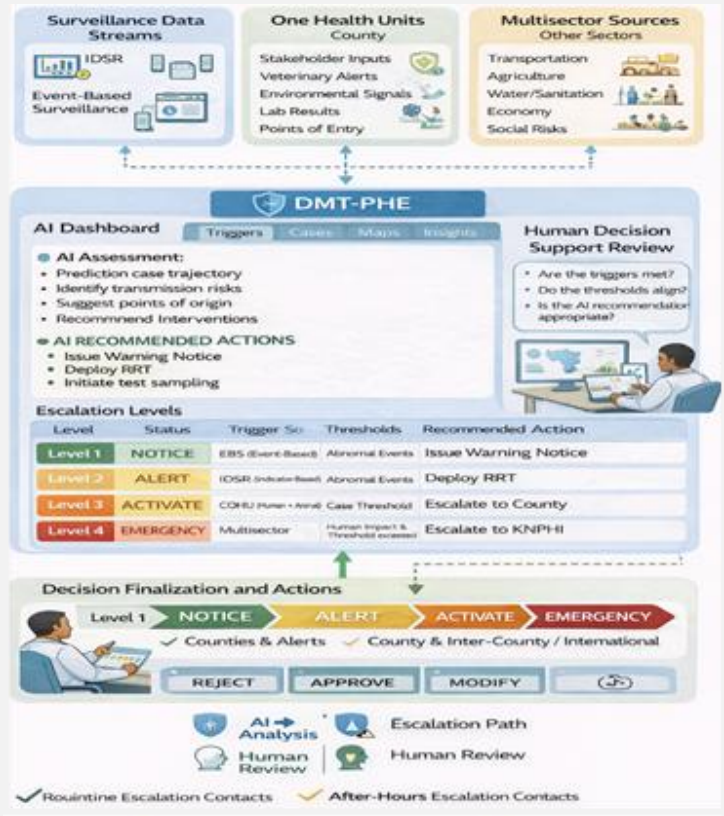


Figure 2: AI-enabled DMT-PHE Agentic architecture, multi-stream input interface, and escalation workflow

Phase 0 (1-6 months): Early minimum viable product (MVP) milestones: Within the first six months, the consortium will deliver and demonstrate a minimum viable RADAR product comprising: (i) an agreed minimum event record for priority surveillance events; (ii) mapped and functioning initial interoperability/data flows for selected priority data sources; and (iii) an integrated DMT-PHE decision-support workflow demonstrating signal ingestion, classification, escalation, and human confirmation, (iv) Initial technical specifications and interface requirements for RADAR/DMT-PHE–GHEC-AI integration, jointly defined with Sobek team during the MVP phase.

Phase 0 Gated Decision: The 4-component MVP will provide an early technical and operational go/no-go point before development or national/county deployment.

Phase 1 (6 months- 1 year): Following successful MVP review, Phase 1 will complete RADAR platform preparation, dataset inventory and mapping, priority system connectors, outbreak user guides, and national DMT-PHE deployment plan, culminating in an integrated platform and parallel validation deployment for RADAR, DMT-PHE and GHEC-AI integration. The GHEC-AI integration interface will be developed, tested, and incorporated into MVP deployment, including validation of the escalation-to-response handoff. During the MVP phase, WSU, UM/CGHE, KNPHI/DHA and the enterprise software-development partner will jointly finalize the TPP and technical architecture, including deployment requirements, interoperability interfaces, security controls, and the RADAR–DMT-PHE–GHEC-AI workflow. Phase 1 concludes with an integrated data platform prototype plus an eight-week parallel validation deployment of DMT, generating a concordance dataset.

Phase 1 Gated Decision: Fully developed RADAR and DMT-PHE prototypes, with all connectors, finalized guides, dataset inventories, deployment and evaluation plan, plus GHEC-AI integration prototype.

Phase 2 (Year 1-2): Deployment of RADAR and DMT-PHE, plus parallel validation deployment at national and five priority county EOCs. Site readiness assessments, ICT setup, staff training, county-level parallel validation, and AI model retraining using national EOC operational data. Following MVP review, the integrated product will undergo product hardening, scalability and load testing, cybersecurity testing, interface validation, and initial deployment within KNPHI/DHA-approved infrastructure.

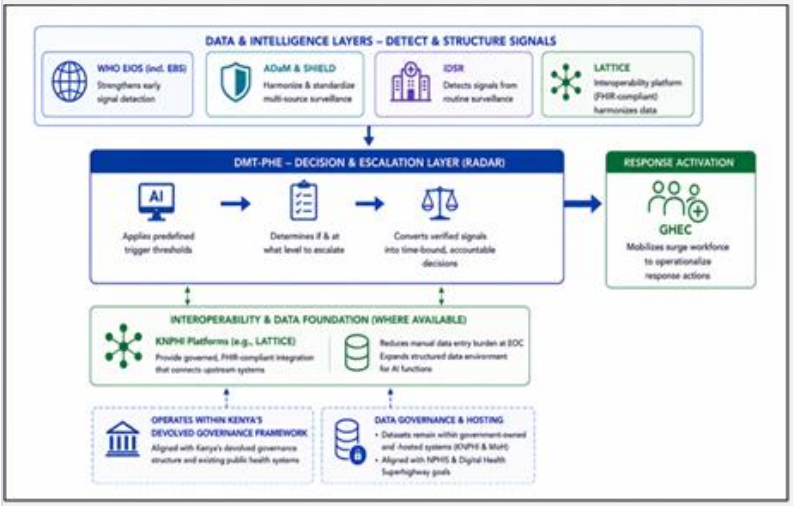


Figure 3: How RADAR will fit the surveillance and response workflow at the KNPHI. IDSR and EIOS detect signals, RADAR flags outbreaks while ADaM structures and manages outbreak data, the DMT-PHE governs escalation decisions, and GHEC mobilizes responders, creating a coherent pathway from detection to coordinated action. GHEC-AI represents the downstream response-matching layer, receiving only authorized, minimum-necessary operational information from the RADAR/DMT-PHE workflow

Phase 2 Gated Decision: >80% dataset integration from baseline, AI-expert concordance ≥0.85 across sites; SUS score ≥75; documentation completeness ≥80%; zero adverse decision events, minimum event record in place and published, and measurable reduction in detection-to-classification time measured by the 7-1-7 timelines; and SUS ≥78 at all sites.

Phase 3 (Year 2-3): Prospective follow up deployment and product evaluation, endline assessment, scale-up evidence reports and policy brief, KNPHI operational transition, and regional dissemination through Africa CDC and WHO. GHEC-AI integration will be evaluated for interoperability, workflow fit, timeliness, usability, and contribution to response mobilization.

Phase 3 Gated Decision: AI-expert concordance ≥0.90; SUS ≥80; false-negative rate below pre-specified ceiling; documented KNPHI recurrent budget pathway; two manuscripts documenting the development and/or deployment.

How RADAR will fit in the surveillance and response landscape in Kenya

The RADAR architecture is designed to complement, not replace, existing surveillance infrastructure in Kenya. WHO EIOS, which includes EBS, strengthens early signal detection. ADaM and SHIELD harmonize and standardize multi-source surveillance data for outbreak management. The DMT-PHE sits above these data and intelligence layers, applying predefined trigger thresholds to determine when escalation is required and at what level, converting verified signals into time-bound, accountable decisions within Kenya's devolved governance framework.

The response activation and surge personnel management platform e.g. the GHEC-AI- integration will form a core downstream component of the RADAR architecture **(Fig. 3)**. RADAR is expected to provide upstream surveillance intelligence, the DMT-PHE AI Agent will support human-supervised classification and escalation, while GHEC-AI will support matching of approved response needs to available workforce and response capacity. Following authorized escalation, only minimum-necessary operational information will be transmitted through secure API/MCP interfaces. In summary, IDSR and EIOS detect signals, RADAR flags outbreaks while ADaM structures and manages outbreak data, the DMT-PHE governs escalation decisions, and GHEC mobilizes responders, creating a coherent pathway from detection to coordinated action. Where available, KNPHI-supported interoperability platforms - including LATTICE - will contribute governed, FHIR-compliant integration that connects upstream surveillance platforms, reducing the manual data entry burden at the EOC level and expanding the structured data environment available to RADAR's AI functions.

Budget – completed by the grantee

❖ In this section, please describe the direct and indirect costs, and total requested grant amount.

The major cost drivers are personnel and fringe benefits ($308,941 across three years), consultants ($157,838), subawards ($800,000 cumulative), and technology infrastructure, including the enterprise server and hosting costs. These investments directly support the design, development, deployment, validation, evaluation, and institutionalization of the RADAR platform and AI-enabled DMT-PHE within KNPHI and county Emergency Operations Centers. Personnel provide scientific leadership, technical development, project coordination, data management, financial oversight, and implementation monitoring, while consultants and subaward partners contribute specialized expertise in AI development, surveillance governance, implementation science, monitoring and evaluation, One Health integration, and emergency operations. Costs are aligned with phased project objectives. Period 1 focuses on platform design, governance, infrastructure acquisition, and implementation planning; Period 2 supports deployment, software optimization, training, and validation under operational conditions; and Period 3 supports evaluation, dissemination, sustainability planning, and transition to KNPHI ownership. Potential spending risks include recruitment delays for key project staff, procurement lead times for ICT infrastructure, slower-than-anticipated stakeholder adoption, and technical integration challenges between surveillance systems. These risks will be mitigated throug h early recruitment, phased procurement planning, strong project managem ent oversight, engagement of experienced technical

partners, regular monitoring, and structured stakeholder engagement with KNPHI and county EOCs. These mitigation approaches are embedded in the project's coordination, governance, and monitoring activities.

Personnel and Benefits $308,941

Personnel: Personnel costs support the design, deployment, governance, implementation, monitoring, and operationalization of the RADAR platform and AI-enabled DMT-PHE within the Kenya National Public Health Institute (KNPHI) Emergency Operations Center (EOC). The requested level of effort reflects only the portion of time directly devoted to project implementation and deliverables.

The project team includes:

Dr. Isaac Ngere (Principal Investigator): Person Months: 1.8 PM (15% effort × 12 months) - Dr. Ngere will provide overall scientific and technical leadership for RADAR implementation, including oversight of AI-enabled outbreak detection workflows, surveillance system interoperability, stakeholder coordination with KNPHI and county EOCs, validation of system outputs, and strategic alignment with national outbreak preparedness priorities. His leadership is required to ensure that the RADAR platform achieves the project objectives of reducing detection-to-escalation timelines and improving outbreak decision-making accuracy. Provides overall scientific, technical, and strategic leadership, oversight of AI-enabled outbreak detection workflows, stakeholder engagement, validation of system outputs, and evaluation of project performance.

Selestine Njoga (Director, East Africa Finance and Operations): Person Months: 0.84PM (7% effort × 12 months). Ms. Njoga will work with the PI and will be directly involved with the National and County Government Units during Stakeholder engagements. Ms. Njoga will also participate in Community

engagement and participate in planning the Inception meetings and further in the Monitoring and Evaluation of the Project to ensure that the Project deliverables are on course. This role is necessary for ensuring proper stakeholder engagement and milestones and that the project expenditures are implemented in accordance with Gates Foundation requirements and organizational policies while maintaining uninterrupted project execution.

Project Coordinator (TBN): Person Months: 8.4 PM (70% effort × 12 months) - The Project Coordinator will serve as the primary operational lead responsible for coordinating implementation activities across KNPHI, county EOCs, technical teams, consultants, sub-contractors and other stakeholders. This role will have managerial responsibilities including activity planning, supervision, milestone tracking, documentation, logistics management, monitoring deliverables, organizing training and stakeholder workshops, and supporting Gates Foundation reporting requirements. Given the complexity of deploying an AI-enabled national surveillance and decision-support platform, this position is central to successful execution of the project. Responsible for day-to-day project management, stakeholder coordination, implementation oversight, milestone tracking, logistics, reporting, and training activities. Recruitment will occur early in the project to support start-up and deployment activities.

Kadenge Wa Ya (Senior Software Engineer): Person Months: 4.2 PM (35% effort × 12 months). Mr. Wa Ya is an experienced software developer who will lead all platform development for RADAR and support the development and integration RADAR, DMT-PHE and other surveillance and response products. This role will strengthen alignment between technical development activities and KNPHI operational requirements, ensuring that project outputs are practical, scalable, and sustainable. Leads software development, systems integration, platform optimization, interoperability, and technical support.

Data Analyst(s) (TBN): Person Months: 6.0 PM (50% effort × 12 months) - The Data Analyst will support data harmonization, quality assurance, dashboard development, analytics generation, monitoring indicators, and validation of surveillance outputs generated through RADAR. The Data Analyst will also support multi-stream surveillance data processing, outbreak analytics, monitoring of AI-assisted classifications, performance reporting, and generation of implementation evidence needed to assess project effectiveness to strengthen the project's ability to measure improvements in outbreak detection, escalation timelines, and decision-support performance. Support data quality assurance, analytics, dashboard development, AI-output validation, monitoring, reporting, and evaluation activities.

Benefits

Fringe benefits are based on approved organizational rates and include employer-paid medical insurance, pension contributions, statutory social security obligations, leave benefits, workers' compensation, and other personnel-related benefits required under organizational policy and Kenyan labor regulations. The fringe costs ensure retention of qualified personnel necessary to implement the highly specialized technical, surveillance, analytics, and operational activities proposed under this project.

Travel - $31,797

Travel costs are essential to support the phased implementation, deployment, monitoring, evaluation, and institutionalization of the RADAR platform and AI-enabled DMT-PHE across KNPHI and participating county Emergency Operations Centers (EOCs). Travel activities will facilitate stakeholder engagement, baseline and operational assessments, system deployment, training, technical support, supervision, monitoring, dissemination of findings, and sustainability planning. County visits are required to assess operational readiness, document surveillance workflows, identify system integration requirements, validate implementation progress, provide technical assistance, troubleshoot challenges, gather user feedback, and strengthen adoption of the platform within routine public health operations. These field engagements are critical to ensuring that RADAR is effectively

integrated into Kenya's disease surveillance and outbreak response architecture and delivers measurable improvements in outbreak detection, classification, escalation, and decision-making. Funds are also requested to support project inception, annual review, and stakeholder coordination meetings involving KNPHI leadership, county EOC representatives, technical partners, and project staff. These meetings provide a structured forum to review implementation progress, validate performance indicators, assess project achievements, address operational challenges, coordinate corrective actions, review lessons learned, and guide system optimization, sustainability planning, and future scale-up. Regular engagement of national and county stakeholders is essential to sustaining ownership, strengthening governance, and ensuring successful institutionalization of the platform.

Travel includes site supervisory visits to participating implementation sites, annual review meetings, stakeholder workshops, and dissemination activities. In addition, conference participation will support presentation of project findings, publication and dissemination of implementation evidence, sharing of lessons learned related to AI-enabled outbreak surveillance, interoperability, and decision-support systems, and engagement with the broader public health and digital health communities. Travel also includes conference funds to support travel to two international meetings. These activities will contribute to knowledge exchange, promote adoption of best practices, and generate visibility for innovative approaches to strengthening public health emergency preparedness and response. All travel costs have been developed using reasonable assumptions regarding the number of trips, personnel, lodging, per diem requirements, and local transportation needs. Road travel will utilize pooled transportation whenever feasible to maximize cost efficiency, while any air travel will comply with organizational and donor travel policies and cost guidelines. These measures will ensure prudent use of project resources while maintaining the level of engagement necessary to achieve project objectives and target outcomes

Consultants $157,838

Consultants provide specialized expertise not available through project personnel. The project requires specialized expertise from KNPHI and monitoring specialists to ensure that RADAR is implemented within existing government governance frameworks and aligned with national public health priorities.

KNPHI Data Governance Consultants: These consultants will provide subject matter expertise on surveillance governance, national data-access, interoperability requirements, data-sharing frameworks, and public health information management standards. Their involvement is necessary to ensure that RADAR complies with KNPHI, Ministry of Health, Digital Health Agency, and One Health governance requirements while facilitating integration of multiple surveillance systems.

Robert Osoti (Monitoring, Evaluation and Learning Consultant): Mr. Osoti will lead implementation monitoring, performance tracking, data quality reviews, and evaluation support. This role is required for measuring achievement of key project outcomes, including reduction in detection-to-classification timelines, improved outbreak decision-making, increased 7-1-7 compliance, and evaluation of user adoption and performance of the AI-enabled decision-support system.

Enterprise Software Consultant Will provide highly specialized expertise for final software enhancements, scalability testing, interoperability optimization, hosting architecture review, cybersecurity strengthening, and preparation of the platform for sustained operational use beyond the grant period. This expertise is necessary to ensuring that RADAR and the DMT is market ready from initial development, remains robust, scalable, secure, and capable of supporting future expansion.

Dell PowerEdge R770 Enterprise Server ($36,095)

The server represents the project's primary technology investment and is required to host the RADAR development platform, integrated surveillance databases, analytics engines, AI services, decision-support tools, and secure repositories for processing and storing surveillance information. The server provides the computing capacity, processing speed, security controls, backup capability, and scalability needed to expand existing computing capacity and support integration of multiple surveillance data streams and execution of AI-enabled outbreak analytics. Without this infrastructure, deployment of a secure, interoperable, and nationally scalable surveillance platform would not be feasible with the existing systems. The equipment investment directly supports the project's objective of creating a deployable and sustainable AI-driven public health surveillance and decision-support platform within KNPHI's EOC environment

More details in the attached budget narrative.

Other Direct Costs Period 1 – 3: $74,025

Direct Costs Period 1: $32,153

ICT supplies include laptops, computers, monitors, accessories, and connectivity equipment required for project staff, technical teams, and KNPHI implementation personnel. These devices will support system administration, software testing, analytics, training, monitoring, and operational use of RADAR and DMT-PHE applications.

Document Development and Training Materials ($4,013)

Office and training supplies will support stakeholder engagements, requirements gathering workshops, user training sessions, technical documentation, validation exercises, and implementation planning meetings. These materials facilitate effective engagement and adoption of RADAR by national and county public health stakeholders.

Stakeholder Engagements ($300)

Funds are requested to support transportation costs for two stakeholder engagement activities involving national and county-

level partners. These engagements are essential for coordinating project implementation, sharing progress updates, obtaining technical input, and ensuring alignment with public health priorities and project objectives.

Community Engagement and Risk Communication ($720)

Funds are requested to facilitate community engagement and risk communication activities that will support effective dissemination of project information and encourage stakeholder participation at the community level. Lunch and transport reimbursements for 20 participants will enable attendance during engagement sessions and focus group discussions. In addition, fuel costs are included to facilitate field-based outreach and project-related travel using county vehicles, ensuring timely access to target communities and implementation sites.

Communication Costs for National, County, and Sub-County Level Staff ($1,200)

Funds are requested for airtime reimbursements for project personnel involved in coordination, data collection, stakeholder follow-up, and routine communication activities. Reliable communication is critical for maintaining engagement with field teams, coordinating implementation activities across multiple locations, and supporting timely reporting and troubleshooting throughout the project period.

Casual Staff ($6,720)

Funds are requested to engage eight temporary casual staff who will provide short-term operational support for project activities. Responsibilities may include participant mobilization, data collection support, data entry, logistics coordination, and administrative assistance during field implementation. Utilizing casual staff provides the flexibility required to meet project demands while ensuring efficient delivery of planned activities.

Ethical Approval ($1,000)

Funds are requested to cover Institutional Review Board (IRB) and other ethical review fees required for the conduct of project-related research activities. Ethical approval is a mandatory requirement before initiating study implementation and ensures compliance with national and institutional research regulations.

Meeting Costs ($300)

Funds are requested to support a project conference or coordination meeting involving key stakeholders and project personnel. The meeting will provide a platform for reviewing progress, discussing implementation challenges, strengthening collaboration, and aligning activities to achieve project objectives.

Hosting Subscription Costs ($4,000)

Funds are requested for hosting services required to support project-related digital platforms, databases, websites, dashboards, or other electronic systems used for data management and information sharing. These services are necessary to ensure secure, reliable, and continuous access to project information and resources.

Internet Connectivity for KNPHI ($1,900)

Funds are requested to support Starlink internet connectivity and associated subscription costs for KNPHI facilities involved in project implementation. Reliable internet access will facilitate communication, virtual collaboration, data transmission, access to cloud-based systems, and timely reporting, particularly in locations where conventional connectivity options may be limited or unreliable.

Direct Cost Period 2: $18,244

Document Development and Training Supplies ($950)

Funds will support document development, training materials, workshop supplies, documentation materials, and other consumables required for county deployments, user training sessions, stakeholder consultations, and operational meetings.

ICT Supplies ($1,746)

Funds will support acquisition of supplemental ICT accessories and technical equipment needed for deployment support, user onboarding, troubleshooting, system monitoring, and integration activities at participating implementation sites.

Annual Partners Meeting ($1,698)

Supports a structured review meeting bringing together project stakeholders, KNPHI representatives, county participants, and technical partners to evaluate progress and coordinate implementation priorities.

Airtime Support ($2,400)

Provides communication resources for project staff and participating implementation teams to facilitate real-time coordination, troubleshooting, monitoring, and technical support across participating sites.

Motor Vehicle Fuel ($2,400)

Supports transportation required for deployment, supervision, monitoring, stakeholder engagement, and technical support

activities throughout implementation counties.

Motor Vehicle Running Costs ($1,200)

Supports servicing, maintenance, and operational costs associated with vehicles used to conduct project-related field activities.

Motor Vehicle Leasing ($600)

Provides temporary vehicle support for implementation activities in situations where project mobility requirements exceed existing transportation capacity.

Internet Connectivity ($2,850)

Supports Starlink subscriptions and internet connectivity required to maintain reliable access to RADAR, facilitate interoperability with surveillance systems, enable remote troubleshooting, support cloud-based services, and ensure uninterrupted access to AI-enabled decision-support tools across implementation sites.

Participant Reimbursements ($4,800)

Supports transportation and participation costs for county EOC personnel, surveillance officers, technical stakeholders, and workshop participants engaged in deployment activities, training sessions, validation exercises, and implementation reviews. These reimbursements are necessary to ensure adequate participation in project activities and successful adoption of RADAR at the county level.

Direct Costs Period 3: $23,628

Office and Administrative Supplies ($1,093)

Funds will support routine administrative materials, printing, documentation, workshop materials, and meeting supplies required for evaluation activities, training refreshers, stakeholder consultations, and dissemination events.

ICT Supplies ($4,000)

Funds will support replacement or supplemental ICT equipment required for system administration, monitoring, analytics, and continued operation of RADAR within KNPHI and participating implementation sites. These resources help ensure uninterrupted access to the platform during the evaluation and transition phase.

Project Dissemination Activities with National and County Stakeholders ($2,120)

Funds will support dissemination meetings to share project findings, implementation experiences, lessons learned, and recommendations with national and county stakeholders. Conference services ($1,000) will cover meeting venue and facilitation costs. Participant support costs ($320) will provide modest lunch and transportation reimbursements to facilitate attendance by key stakeholders. Fuel support for county vehicles ($800) will enable travel for county representatives and project teams participating in dissemination and stakeholder engagement activities.

Airtime ($1,200)

Funds will support monthly airtime top-up cards for project staff and key implementation personnel to facilitate routine communication, coordination, stakeholder engagement, technical support, and follow-up activities across participating counties. Reliable communication is critical for maintaining project operations, coordinating surveillance activities, and supporting implementation partners.

Motor Vehicle Fuel ($645)

Funds will support local transportation requirements associated with project implementation, monitoring visits, stakeholder engagement, supervision activities, and coordination meetings. Vehicle fuel is necessary to facilitate timely field-based activities and maintain engagement with county and national partners.

Research Assistant Costs ($6,720)

Funds are requested to support eight casual research assistants who will provide field-based technical and operational support for project implementation. The research assistants will assist with participant mobilization, community engagement activities, data collection, data quality checks, documentation, and other project-related field activities. Their involvement is essential to ensure timely execution of study activities, adherence to project protocols, and the collection of high-quality data across implementation sites. The use of short-term research assistants provides the necessary surge capacity to meet project requirements while maintaining cost efficiency.

Participant Reimbursements ($2,000)

Funds will support participant reimbursements associated with evaluation activities, stakeholder consultations, interviews, workshops, and other project-related engagements. These reimbursements will facilitate participation by key informants and stakeholders whose input is necessary for project assessment, documentation of outcomes, and sustainability planning.

Internet Connectivity ($2,850)

Funds will support Starlink internet subscriptions required to maintain reliable connectivity at project implementation sites. Continuous internet access is essential for data transmission, system access, remote technical assistance, platform monitoring, and interoperability functions that support operation of the RADAR system.

Annual Partners Meeting ($1,000)

Funds will support an annual project review meeting involving KNPHI, WSU, county representatives, technical partners, and other key stakeholders. The meeting will provide a forum to review project achievements, discuss evaluation findings, identify sustainability priorities, document lessons learned, and plan for long-term institutionalization and future scale-up of the surveillance platform.

Annual Hosting Costs ($4,000)

Hosting services remain critical to ensure uninterrupted operation of the RADAR platform during the final project period and transition phase. Funds will support cloud hosting infrastructure, data storage, system maintenance, security services, backup systems, and technical support. Continued hosting will enable ongoing platform availability, support prospective evaluation activities, facilitate operational handover to KNPHI, and promote long-term sustainability of the AI-enabled disease surveillance infrastructure beyond the award period.

Geography Served – completed by the grantee

- List all countries, sub-regions, and/or states where this work will have impact. For additional information, see the Geography and Location of Work – Frequently Asked Questions.

Geography Served	Amount
Kenya	$1,500,000

Location(s) of Work – completed by the grantee

- List all countries, sub-regions, and/or states where this work would be performed. The light touch process does not allow locations of work in India and/or China. For additional information, see the Geography and Location of Work – Frequently Asked Questions.

Location of Work	Amount
Kenya	$1,500,000

- If World is listed for Location of Work, confirm that you will not use grant funds to support activities in the following countries: India and/or China.

  ☐ Confirmed

Anti-Terrorism and Use of Funds

- Did you list any locations of work that are subject to U.S. government sanctions as identified below?

☒ No

☐ Yes

Comprehensive Sanctioned Countries	List-based Sanctioned Countries
☐ Cuba ☐ Ukraine's Crimea Region or so-called Luhansk or Donetsk People's Republics ☐ Iran ☐ North Korea ☐ Syria	☐ Countries subject to list-based sanctions* **Note: Countries subject to list-based sanctions can be found on the U.S. Department of Treasury website*

Did you list any locations of work as "to be determined" (TBD) or geographic areas instead of specific countries (e.g., Sub-Saharan Africa or Middle East)?

☒ No

☐ Yes

If you selected 'yes' to TBD locations or locations that are not country-specific, and/or selected locations that are subject to U.S. government sanctions or believe that project work could reasonably occur in a location that is subject to U.S. government sanctions, please review the guidance in the Office of Foreign Assets Control (OFAC) Guidelines and respond to both the following questions.

Confirm that you have discussed the location of work with your foundation contact.

☐ Confirmed

☐ Not confirmed (please explain)

Confirm your organization will ensure compliance throughout the term of the project with the anti-terrorism clause in our grant agreement.

☒ Confirmed

☐ Not confirmed (please explain)

Measurement and Evaluation

Describe your plan for monitoring, measurement, and evaluation of the outputs and outcomes. Please specify (a) the learning/evaluation questions for this project and how you plan to answer them through monitoring, measurement, and/or evaluation; (b) the resources (financial, technical, human) that you need to ensure high quality data.

WSU will employ prospective monitoring, evaluation, and learning framework to assess whether RADAR improves the timeliness, consistency, and quality of outbreak decision-making. Key learning questions will include, among others:

1) Does RADAR reduce time from signal detection to DMT-PHE classification and escalation?

2) Does it improve concordance with expert classification, documentation completeness, 7-1-7 performance, and user acceptability?

3) Does integration of multiple surveillance streams provide earlier or additional actionable intelligence, and what technical, workflow, or safety challenges affect adoption and scale-up?

4) The evaluation will also assess the RADAR/DMT-PHE–GHEC-AI handoff, including technical reliability, completeness and timeliness of information exchange, user experience, and whether the integration improves progression from authorized escalation to response mobilization

These questions will be assessed using routinely captured system logs and surveillance indicators, baseline-to-post-deployment comparisons, expert-panel benchmarking, usability assessments, workflow observations, and stakeholder interviews. High-quality measurement will require dedicated MEL financing, secure data-management and analytics infrastructure, and technical expertise in epidemiology, surveillance, AI/model evaluation, implementation science, qualitative research, and data management. WSU will lead this end term evaluation, while KNPHI and county EOC personnel will support routine data capture

and validation, while consortium technical and evaluation teams will conduct quality assurance, periodic performance reviews, and independent verification of key outcomes.

Are you planning a formal evaluation?

☐ No

☒ Yes (please explain)

If yes, describe when the formal evaluation will be conducted during the project, who will conduct it (external/third party or not), the methodology you will consider, and how the main evaluation audiences will use the findings. For additional information, see the foundation's Evaluation Policy.

A formal evaluation will be conducted at **two points**: **a baseline evaluation** at project initiation and an **end-line evaluation** toward project completion. WSU will lead both evaluations, working with KNPHI and other consortium partners; this will therefore be an internally led evaluation rather than an independent third-party assessment, while maintaining predefined methods and transparent analysis. Consistent with the Gates Foundation's fit-for-purpose approach, the evaluation will use mixed quantitative and qualitative methods, including baseline/end-line comparison of surveillance and decision-making indicators, 7-1-7 performance measures, system-use and workflow data, user surveys, and key-informant interviews. Findings will be used by KNPHI and county teams to improve implementation and institutionalization of DMT-PHE, by the consortium to refine and scale RADAR, and by the Gates Foundation and other stakeholders to inform scale up, replication, policy decisions and future investment.

Data Access

It is anticipated that the activities of this project will generate or collect datasets that may be useful to the foundation, project collaborators, third-party researchers, and/or the public. Please answer the questions below with a level of detail appropriate to the project activities.

What types of data and specific datasets will be generated from the project activities and/or collected from other sources?

The RADAR project is expected to generate and/or use several categories of data, including: (1) routine and event-based disease surveillance data from systems such as IDSR/KHIS2, EBS, NPHIIS and related outbreak line lists; (2) laboratory and pathogen surveillance data; (3) One Health data, including animal surveillance data from KABS and relevant environmental, climate and meteorological datasets; (4) historical outbreak, response, and PHEOC operational data used for model development, validation, and benchmarking; (5) derived analytic datasets generated through data integration, harmonization, anomaly detection, risk scoring, forecasting, and decision-support workflows; and (6) implementation and evaluation data, including system performance metrics, alert-to-decision timelines, 7-1-7 indicators, user interactions, usability assessments, model outputs, error/override logs, and qualitative feedback from stakeholders. Where appropriate, de-identified and aggregated datasets, metadata, analytic code, data dictionaries, and model documentation will also be generated to support reproducibility, learning, and future research, subject to applicable ethical, legal, data-sharing, and Government of Kenya governance requirements. Telecommunications, mobility, media-scanning, and other digital data will be used only where legally and operationally permitted and at the minimum granularity necessary for the approved surveillance purpose, with aggregation or de-identification applied where appropriate.

Do you plan to create a data management plan? If not, why not?

Yes.

- What data curation (e.g., organizing, describing, cleaning, enhancing, and preserving) activities are planned to enable reuse or secondary use of the datasets?

RADAR datasets will be curated through standardized data dictionaries and metadata, harmonization of formats and coding across source systems, quality checks for completeness and consistency, de-identification where required, and version-controlled documentation of derived variables and analytic transformations. Curated datasets, code, and supporting documentation will be preserved in secure, governed repositories using interoperable formats to enable appropriate reuse, subject to consent, privacy, data-sharing agreements, and Government of Kenya data-governance requirements. Given the recent requirements by DHA on systems interoperability, we will work closely with KNPHI to ensure our data storage needs

match what the DHA recommends.

The production system will be deployed within KNPHI/DHA-approved on-premise or other Government-authorized infrastructure, while development environments may use approved non-identifiable data. Access will be governed through role-based permissions, least privilege, multifactor authentication, encryption, audit logging, backup/recovery, and incident-response controls, with administration and data-residency requirements defined by KNPHI/DHA

Where and when will the datasets be deposited and preserved to enable reuse or secondary use and under what license?

Datasets containing Government of Kenya surveillance or other restricted health data will be deposited and preserved only within approved Government of Kenya–controlled or authorized infrastructure, in accordance with DHA requirements, the Data Protection Act, and applicable data-sharing agreements. Identifiable or otherwise restricted data will be stored and processed only within Government-controlled or specifically authorized infrastructure and will not be transferred to external institutions, public generative-AI services, or external cloud environments without explicit government authorization, applicable data-use agreements, ethics approval where required, and security review.

During the project, curated datasets will be archived as they are generated, with access limited to authorized users; any secondary use will require the relevant government/data-custodian approvals. Where permissible, de-identified or aggregated derivative datasets, metadata, code, and documentation may be shared under an appropriate open license, while restricted source data will remain under controlled access rather than an open-data license.

Please describe any informed consents, approvals, and/or agreements that may be required to enable use or reuse of the datasets by the foundation, project collaborators, and/or third-party researchers. What steps do you plan to take to obtain such consents, approvals, and agreements?

Use or reuse of RADAR datasets will be subject to applicable ethical approvals, Government of Kenya data-access permissions, data-sharing agreements, and institutional agreements. Where individual-level data are collected specifically for research or evaluation, appropriate informed consent or an approved waiver will be obtained through the relevant ethics review process at KEMRI scientific and ethics review committee. For routine government surveillance data, access and secondary use will be governed by the responsible data custodians, DHA requirements, KNPHI/MOH approvals, and applicable data-sharing agreements. The project will establish these approvals and agreements during project initiation and will require any external collaborator or third-party researcher to obtain the necessary ethical and data-use approvals before access is granted.

PROGRESS NARRATIVE *(if applicable)*

This section is for progress reporting on funded grants. For each required progress report, complete a version of this progress narrative section and submit the Investment Document to your foundation contact by the date indicated in your grant agreement or latest amendment. If you have any questions or need support, please reach out to your foundation contact.

Progress Report Number	**Reporting Period Start Date**	**Reporting Period End Date**	**Progress Report Due Date**	**Scheduled Payment Amount** (if applicable)
	[DD Month YYYY]	[DD Month YYYY]	[DD Month YYYY]	$

Progress Details

- What progress have you made toward achieving the project's desired outcomes? Describe what you did in this reporting period and what work is planned or anticipated for the next period.

FINAL NARRATIVE

For completed grants only. At the end of your grant, complete this final narrative section and submit to your foundation contact by the date indicated in your grant agreement or latest amendment. If you have any questions or need support, please reach out to your foundation contact. Please find contact information under Investment Details.

Grant Start Date	Grant End Date	Final Report Due Date	Unexpended Grant Funds (if applicable)	Remaining Interest or Currency Gains (if applicable)
[DD Month YYYY]	[DD Month YYYY]	[DD Month YYYY]	$	$

Final Results

- Provide information about what you achieved during the grant and your progress towards achieving the project's desired outcomes and outputs.

Grantee Confirmation

By submitting this report, I declare that I am authorized to certify, on behalf of the grantee identified under Investment Details, that I have examined the responses provided and related attachments, and that to the best of my knowledge, they are true, correct, and complete.

I also confirm that the grantee identified under Investment Details has complied with all of the terms and conditions of the grant agreement.

Your Name	Report Submitted Date
	[DD Month YYYY]